

Do Optimizers Choose Different Functions at the Same Loss?

Project plan (9.520, Fall 2026)

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October 5, 2026

1 Problem

Overparameterized networks have many parameter settings with the same training loss, and the optimizer decides which one training returns. In linear models each first-order method converges to the max-margin solution of its own norm: ℓ_2 for gradient descent (Soudry et al., 2018), ℓ_∞ for Adam (Zhang et al., 2024) and the spectral norm for Muon (Fan et al., 2025). For nonlinear networks the evidence is mixed: adaptive methods were found to generalize worse (Wilson et al., 2017), while recent theory finds that Adam learns richer features than gradient descent and can be more robust (Vasudeva et al., 2025). We ask a causal question: *at matched training loss, do SGD, Adam and Muon return functions of different complexity, curvature and robustness, and does switching optimizer move the function toward the new optimizer’s typical solution?*

2 What we plan to achieve

1. **A model bank** of converged networks trained with SGD with momentum, AdamW and Muon, on an image task (CIFAR-10) and a language task (TinyStories) at three model sizes, plus a linear sanity check.
2. **A large-scale function comparison** of these models through input-derivative metrics of first to third order and a broad set of weight norms. The norms test the linear-theory prediction that each optimizer keeps its own norm small, which is exact on separable data, in nonlinear networks and across model sizes.
3. **A causal test by swapping.** If the optimizer selects the function, switching from A to B after convergence should move the metrics toward B ’s values, unless A ’s solution is stable under B . We test all ordered pairs, with B ’s optimizer state carried over in shadow to avoid a cold start.

3 How we plan to achieve it

3.1 Tasks

(0) Linearly separable synthetic data with a linear model, as a check against the known limits. (A) A small GELU convolutional network on CIFAR-10. (B) A small GELU transformer language model on TinyStories, the setting Muon was designed for. Each task is run at three model sizes: input dimension for task 0, width for the CNN, roughly 1M, 3M and 10M parameters for the language model. The architecture is otherwise fixed within each task.

3.2 Matched-loss model bank

At each model size, each optimizer is trained with three learning rates and three seeds (81 runs per task), and trained to convergence, i.e. until the training loss plateaus. We report the final training and test loss with every metric. Steps are recorded, as the function can drift at constant loss.

3.3 Function metrics

Input-derivative metrics. At test points x , let f be the loss at the true label as a function of the input, $H(x) = \nabla^2 f(x)$, $\lambda(x)$ its largest eigenvalue with unit eigenvector $v(x)$, and $T(x) = \nabla^3 f(x)$. For the language model, x is the sequence of input embeddings and f the next-token loss. Each of the metrics below measures how fast f , or one of its derivatives, varies with the input, so a simpler function should have smaller values and less spread across test points; we report the mean and the standard deviation over the test set.

- *First order*: the Frobenius norm of the input Jacobian of the logits, a measure of sensitivity linked to generalization and robustness (Novak et al., 2018).
- *Second order*: the Frobenius norm of H , estimated with Hutchinson’s method from Hessian-vector products, and λ , by power iteration, both linked to robustness by CURE (Moosavi-Dezfooli et al., 2019); and the curvature of the level sets of f , i.e. how fast the gradient direction turns, a projected variant of the normalized curvature of Srinivas et al. (2022). With $\hat{g} = \nabla f / \|\nabla f\|$ and $P = I - \hat{g}\hat{g}^\top$, it is $\|PHP\|_F / \|\nabla f\| = \sqrt{\mathbb{E}_z \|PH(Pz)\|^2} / \|\nabla f\|$, estimated through random probes z with identity covariance.
- *Third order* (novel, to the best of our knowledge): how the curvature changes between nearby points. All three quantities use one primitive, $T[u, u, \cdot] = \nabla_x (u^\top H(x)u)$ with u held fixed, computed by differentiating a Hessian-vector product.
 - *Size of T* : its spectral norm $\|T\|_\sigma = \max_{\|u\|=1} |T[u, u, u]|$, estimated by the shifted tensor power iteration $u \leftarrow (T[u, u, \cdot] + \alpha u) / \|T[u, u, \cdot] + \alpha u\|$ from a few random starts.
 - *Third derivative along the sharpest direction*: $T[v, v, v] = \frac{d^3}{dt^3} f(x + tv)|_{t=0}$, with v from the power iteration on H and its sign fixed by $v^\top \nabla f \geq 0$.
 - *Curvature length scale*: $\lambda / \|\nabla_x \lambda\|$, the distance over which, to first order, the top curvature doubles or vanishes; small values mean the curvature varies sharply. For simple λ , $\nabla_x \lambda = T[v, v, \cdot]$.

Weight norms. The ℓ_0 , ℓ_1 , ℓ_2 , ℓ_3 and ℓ_∞ norms of all weights; per layer, the spectral norm, the nuclear norm and the stable rank; and the path norm, the spectral complexity and the distance from initialization, the last three among the generalization measures compared by Jiang et al. (2020).

Accuracy and robustness. Test accuracy (next-token accuracy for the language model). For the image models, accuracy under Gaussian input noise; for the language model, the sensitivity of Hahn et al. (2021), i.e. how often the prediction changes when a single input token is replaced.

3.4 Swap intervention

We run a swap for every ordered pair (A, B) of distinct optimizers among SGD, AdamW and Muon, six in total. The swap happens after convergence (cf. Keskar and Socher, 2017): from the converged model of A , we continue training with B until it converges again. To avoid a cold start, B ’s state is accumulated in shadow during A ’s run, from A ’s gradients and without applying B ’s updates, so no learning-rate re-warm is needed. The control is A continued with A for the same number of steps. The effect of the swap is the change in each metric relative to this control, compared with the pure- A to pure- B gap.

3.5 Analysis

For each metric, the gap between optimizers in units of the seed-to-seed spread, and the fraction of it a swap closes. A large fraction means the optimizer still shapes the function inside the basin; a small one, that the bias is set earlier.

4 Timeline

- **Oct 6–18**: code for the three optimizers (with shadow state), matched-loss stopping and the metrics; task 0.
- **Oct 19–30**: model bank on task A and first metric comparisons, for the initial checkpoint.
- **Oct 31–Nov 15**: model bank on task B; swaps on task A.
- **Nov 16–30**: swaps on task B; statistics across seeds.
- **Dec 1–4**: presentation. **Dec 5–11**: final paper, public code and Colab notebook.

References

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